1. a)

	0	1	2	3	4	5	6	7
0	0	1	0	0	0	0	0	0
1	0	0	0	1	0	0	0	0
2	0	1	0	0	0	0	0	0
3	0	0	1	0	1	0	0	0
4	0	0	0	0	0	1	0	0
5	0	0	0	0	0	0	0	1
6	0	0	0	0	1	0	0	0
7	0	0	0	0	0	0	1	0

b) Vertex list V: {1, 2, 3, 4, 5, 6, 7}

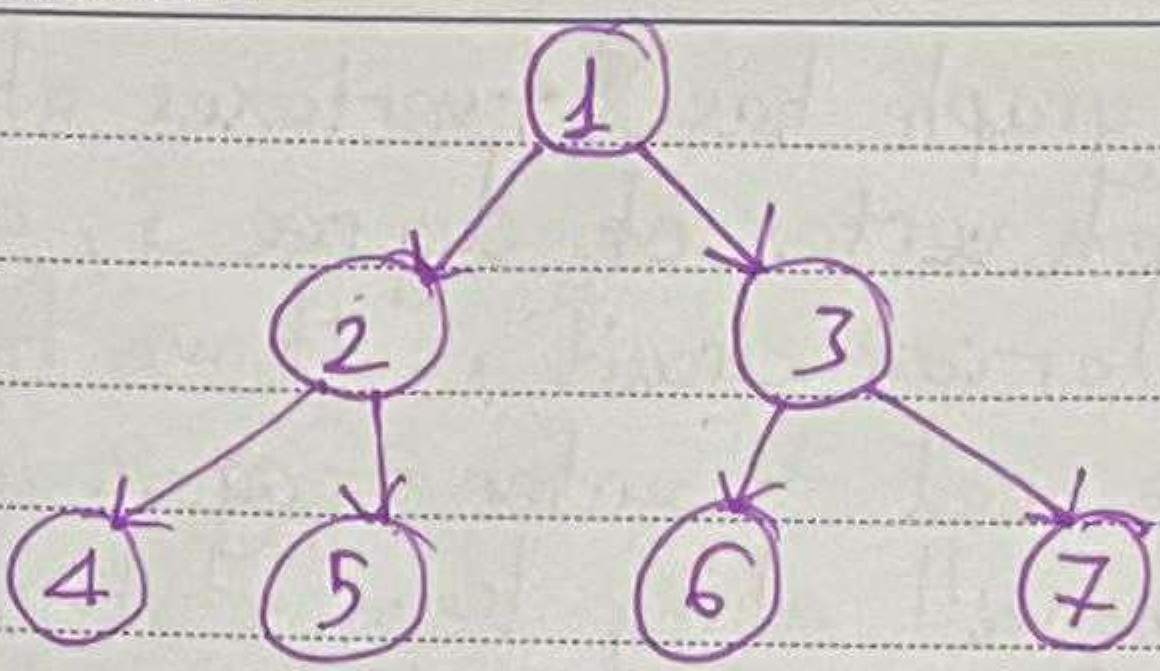
Edge list ~~E~~:

E: {(0,1), (1,3), (2,1), (3,2), (3,4), (4,5), (5,7), (6,4), (7,6)}

c) Adjacency-list:

{0: [1], 1: [3], 2: [1], 3: [2,4], 4: [5], 5: [7], 6: [4], 7: [6]}

2.



Adjacency list : {1: [2,3], 2: [4,5], 3: [6,7],
4: [], 5: [], 6: [], 7: []}

Adjacency matrix:

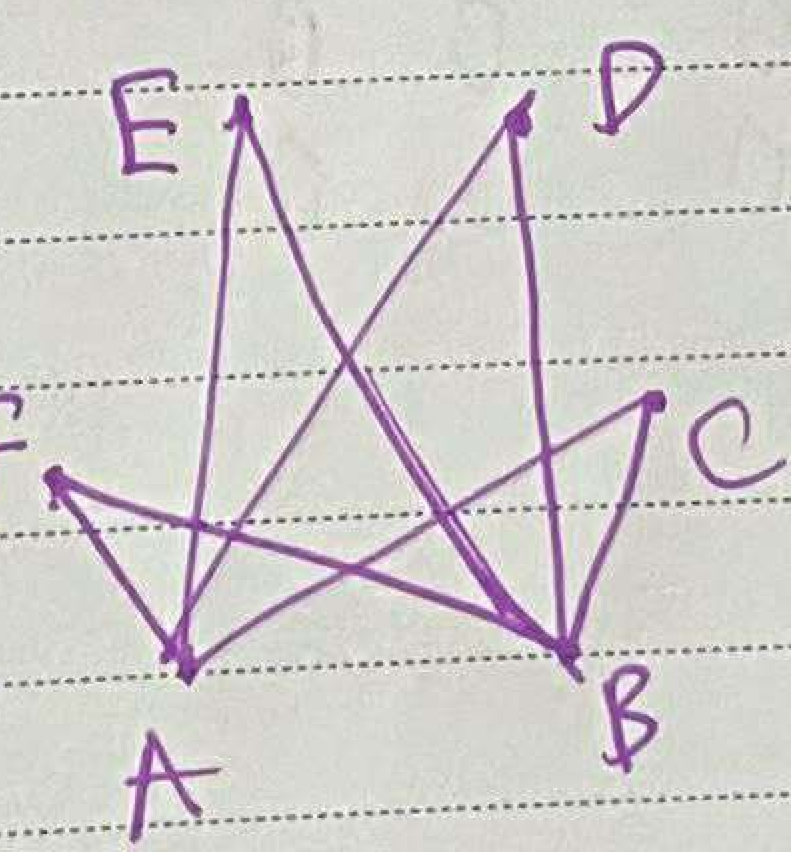
	1	2	3	4	5	6	7
1	0	1	1	0	0	0	0
2	0	0	0	1	1	0	0
3	0	0	0	0	0	1	1
4	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0
6	0	0	0	0	0	0	0
7	0	0	0	0	0	0	0

3. G_1) This graph has ~~8~~ 4 vertexes of degree 3, foreach vertex of degree 3, assume there is an eulerian circuit, if we first leave that vertex, we need to enter it again and leave again, meaning we will alway leave that vertex and vice versa, if we enter that vertex first, we will enter that vertex once again and can not leave. Now we have 4 vertexes of degree 3, ~~meanwhile~~ because we can't start from more than 1 vertex and finish at more than 1 vertex, there is no valid way to preserve the aforemention logic. Thus there is no eulerian circuit here

G2) Same as G1, G2 also has 4 vertexes of degree 3, thus no eulerian circuit

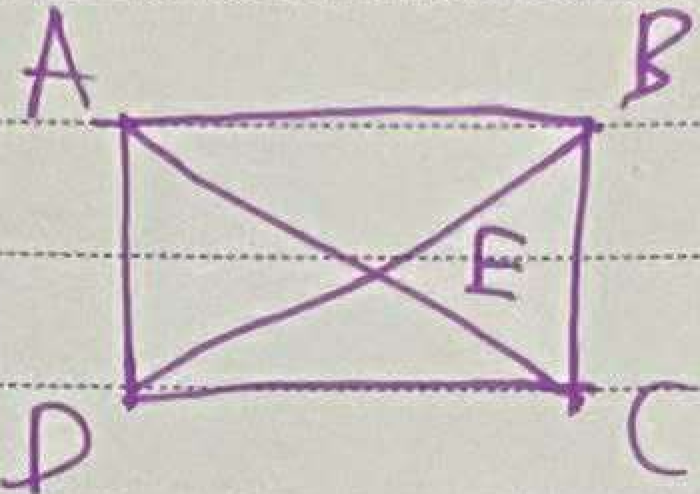
G3) Same as G2, no eulerian circuit

G4)



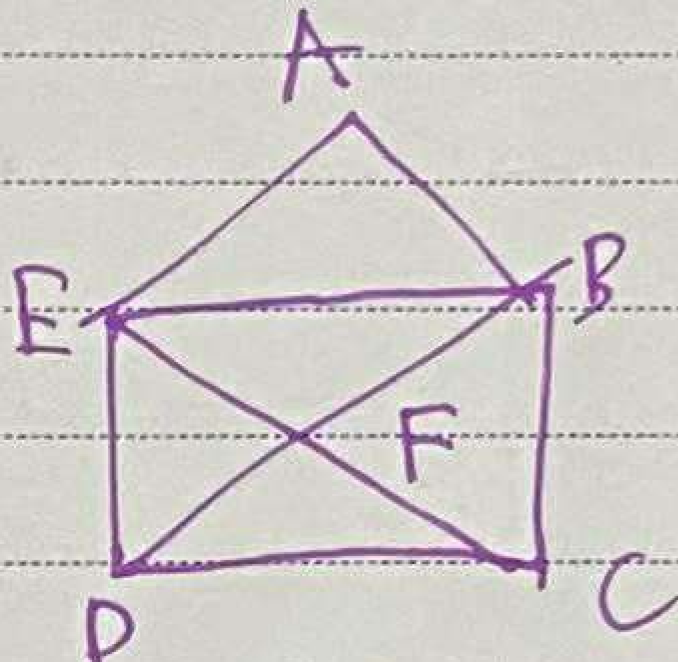
Eulerian circuit: $A \to C \to B \to E \to A \to F \to B \to D \to A$

4. a)



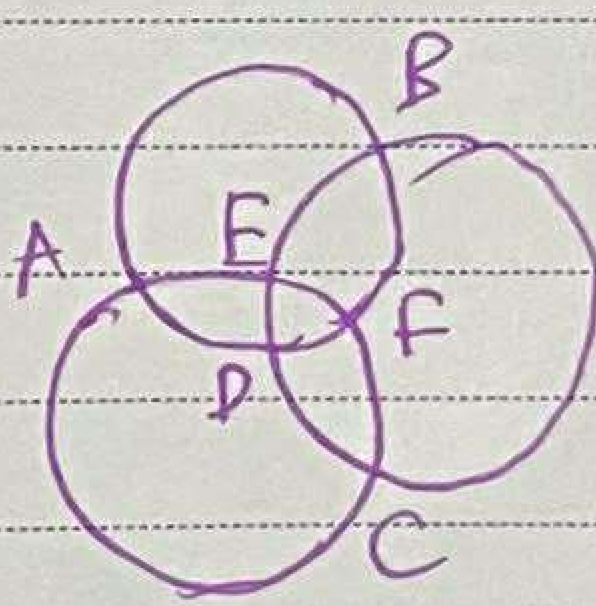
Follow the same logic as part 3, here we have A, B, C and D are degree 3, thus we have to pass them 3 times. But as proven, we can not do such thing, so we can't draw this one without violating the rules.

b)



D → B → A → E → B → C → E → D → C

c)



D → E → F → D → A → E → B → F → C → A → B → C → D